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Properties of Materials
Lecture 15

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CTGU/UoL JEP, September 2024

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Lecture 12: The Physical Basis of Young's Modulus, Learning Objectives, Moduli of Crystals, Rubbers and the Glass Transition Temperature, Composite, Composite Terminology, Fibre-Reinforced Composites, Rule of Mixtures (1)
Lecture 13: Rule of Mixtures (2), Task, Concepts and Terminology (7)
Exercise Seminar 3
Lecture 14: The Yield Strength, Tensile Strength, Hardness and Ductility, Case Studies, Learning Objectives, How do Crystalline Materials Fail (1), Length Scales, Continuum Plastic Flow, Shearing, Shearing of a Ductile Polycrystal, Plastic Instability, What is Creep, Where is Creep Important
Lecture 15: How Does Creep Happen, Lattice Diffusion, Other Diffusion Paths, Creep Resistant Materials, Concepts and Terminology (8), Learning Objectives, How do Crystalline Materials Fail (2), Fracture Concepts, Fracture Jargon, Fracture Length Scales

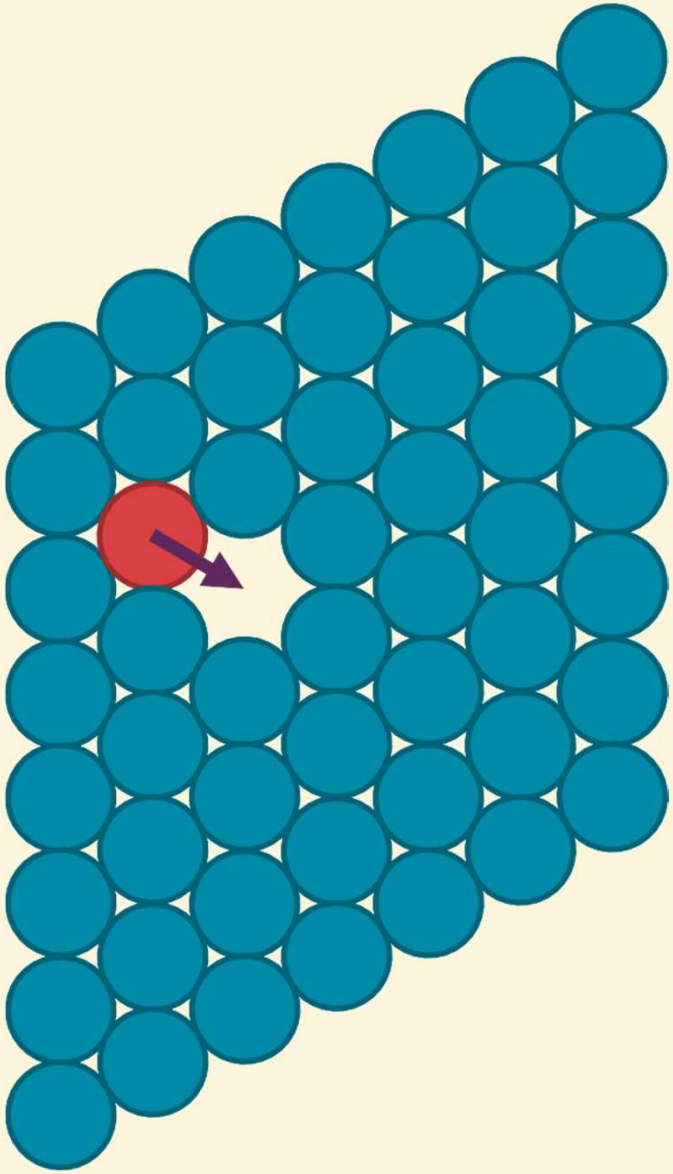
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The yield strength, tensile strength, hardness and ductility

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How does creep happen?

- One of the main mechanisms of creep is by **diffusion**:
  - Movement of atoms through the structure of a material
  - In crystalline materials this is facilitated by **vacancies**



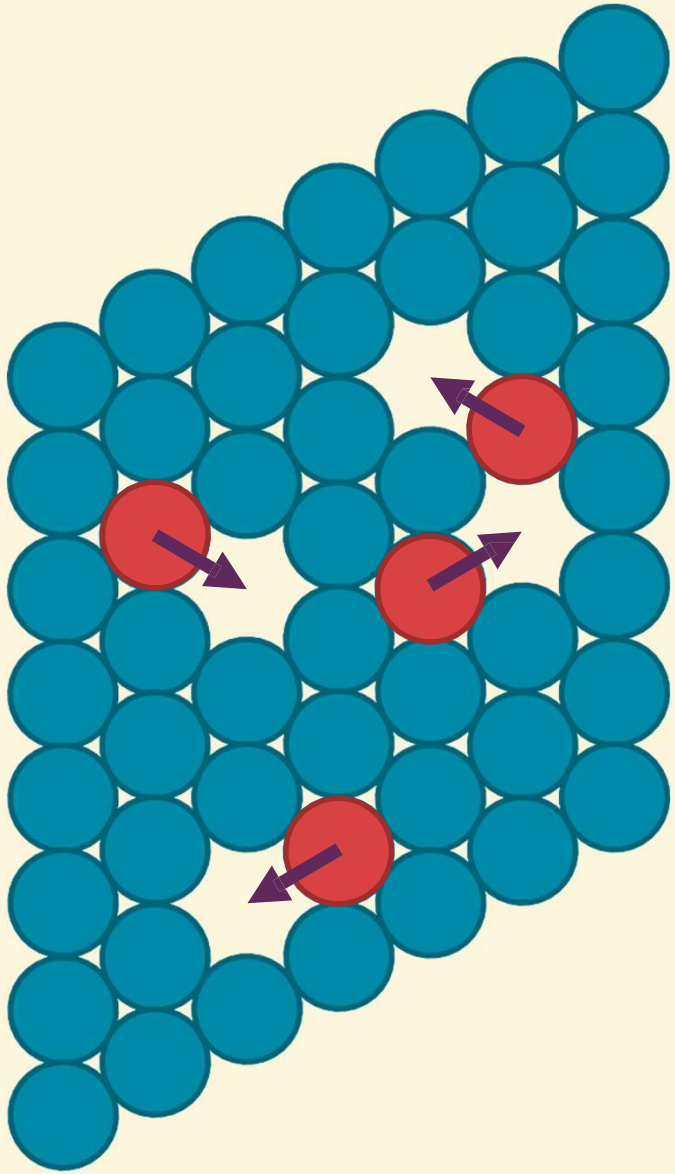
- Each vacancy allows a neighbouring atom to 'jump' between atomic sites
- But atoms **need a little help**…

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Lattice diffusion (1)

- **Thermal energy** allows atoms to move via vacancies
- Under **no load**, movement is **completely random**
- **No overall change**

High temperature

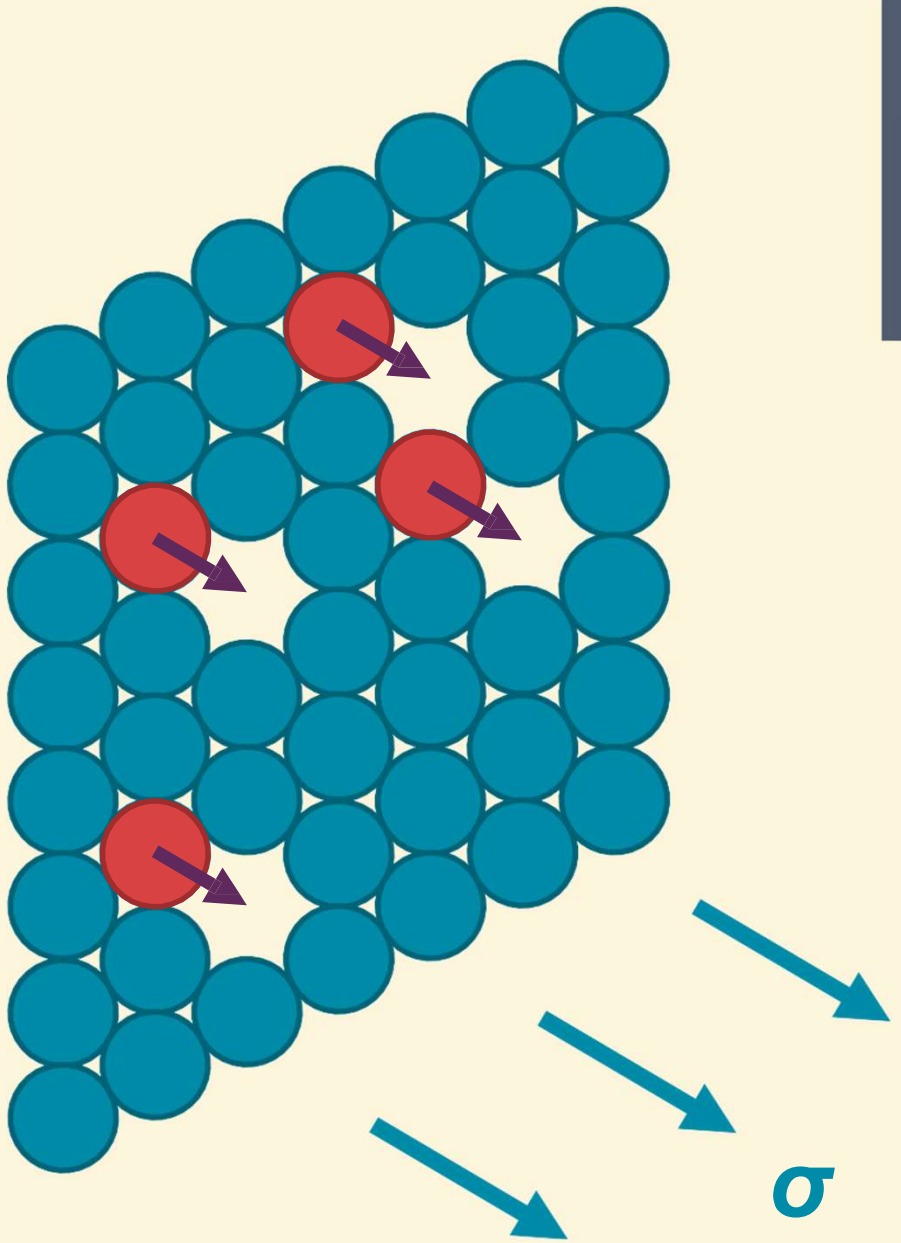


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Lattice diffusion (2)

- **Applied loads** add **directionality** to atomic movements
- Creates a **flux** of mass and vacancies
- Results in **deformation**

High temperature

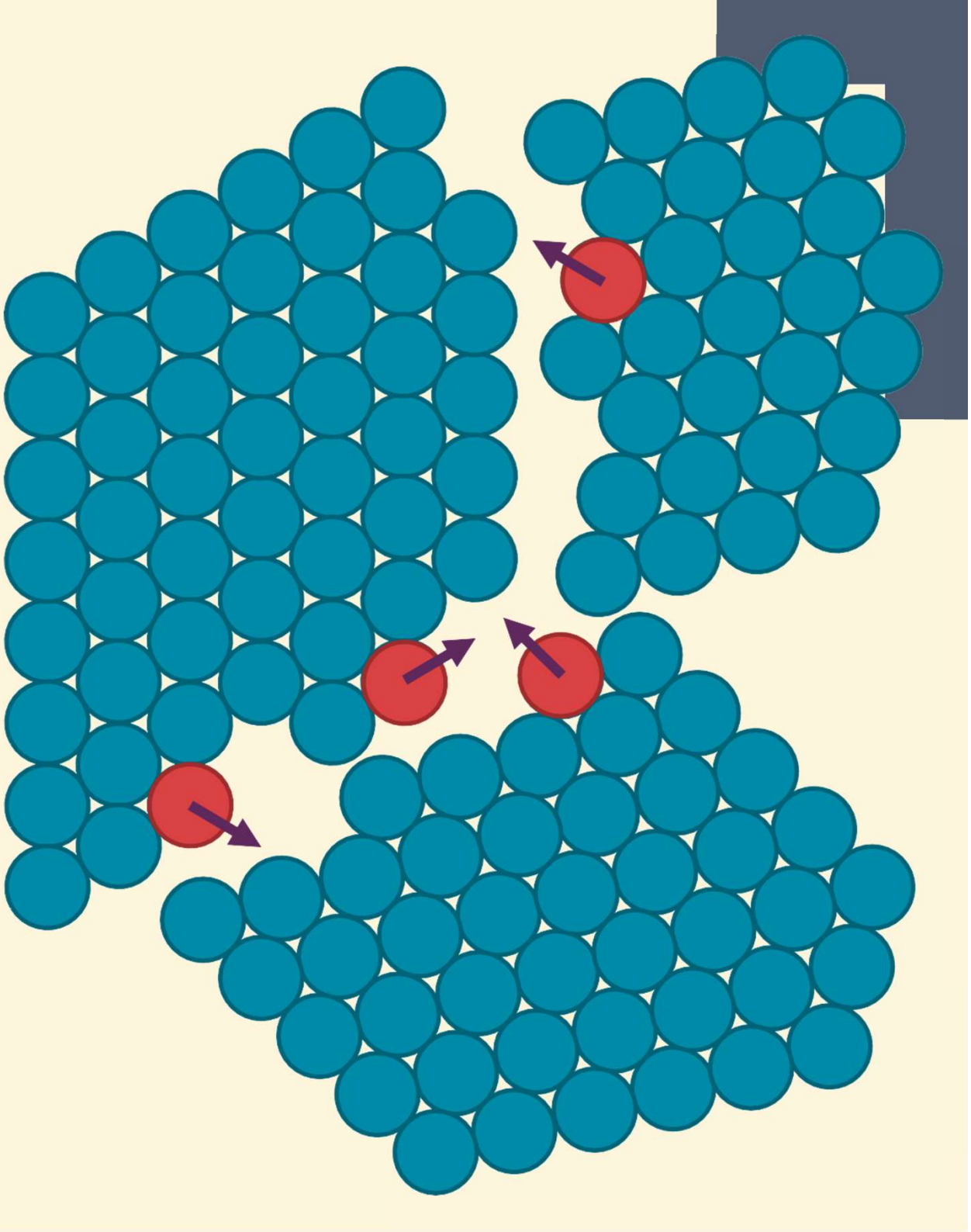


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Other diffusion paths

- Where else might diffusion occur in polycrystalline materials?

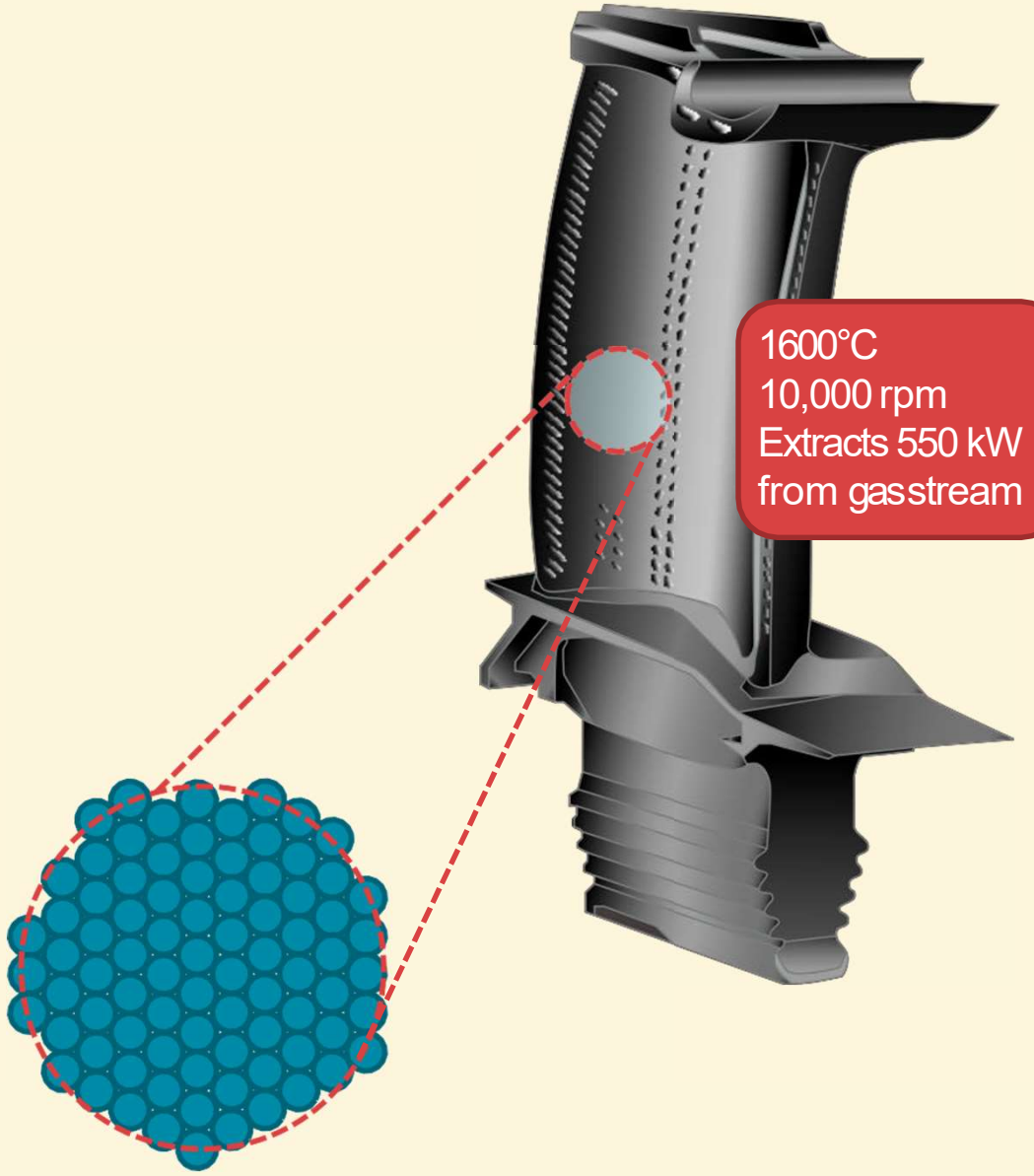
- **Grain boundaries**!



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Creep resistant materials

- What applications require creep resistance?
- What kinds of materials can resist creep?



High-pressure turbine blade
https://commons.wikimedia.org/wiki/File:GaTurbineBlade.svg

Summary

- It is important to consider **length scales** when discussing material properties and behaviour:
  - Being able to **link microscopic and macroscopic behaviour** is key
- **Plastic flow occurs by shear** in polycrystalline materials, with the maximum stress acting at 45° to an applied tensile load
- At the onset of yielding, materials behave in one of two ways:
  - Increasing in strength, due to **strain hardening**
  - By displaying **plastic instability** (i.e. necking)
- **Creep** is the deformation that occurs at **high temperatures** under relatively **low applied stresses** – one of the key mechanisms by which it occurs is **diffusion**

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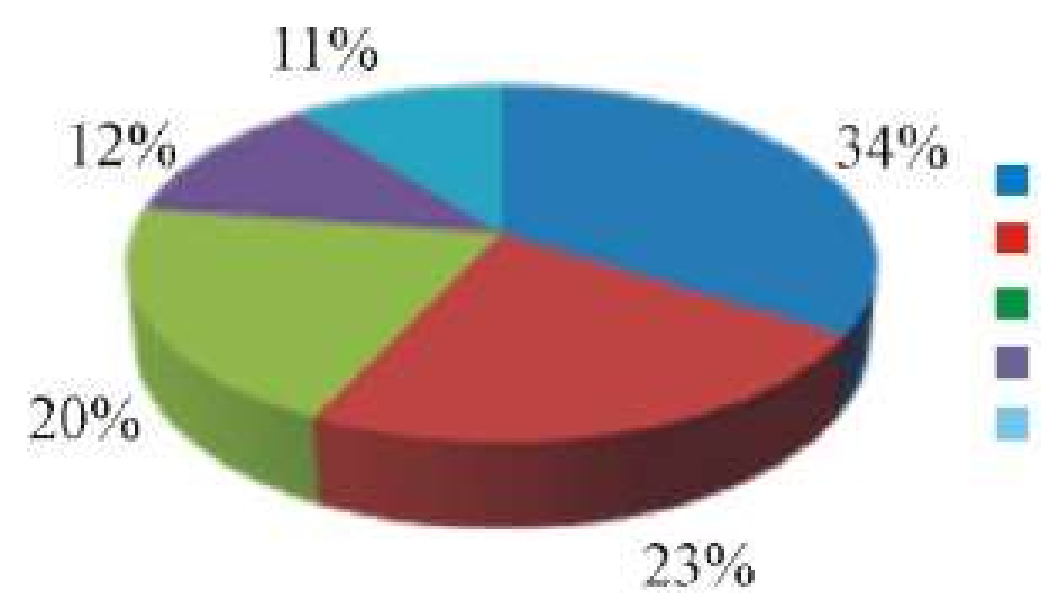
Quality problems in transmission line equipment

- The transmission line equipment, which are the important parts of the power system, play a crucial role on the safe and stable operation of transmission lines. Failures occur at locations such as steel towers, wires, fittings and insulators.

- The quality problems derive from four aspects:
  - Transmission tower
  - Wire
  - Fittings
  - Insulator

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Types of ***transmission tower*** problems proportionally distributed



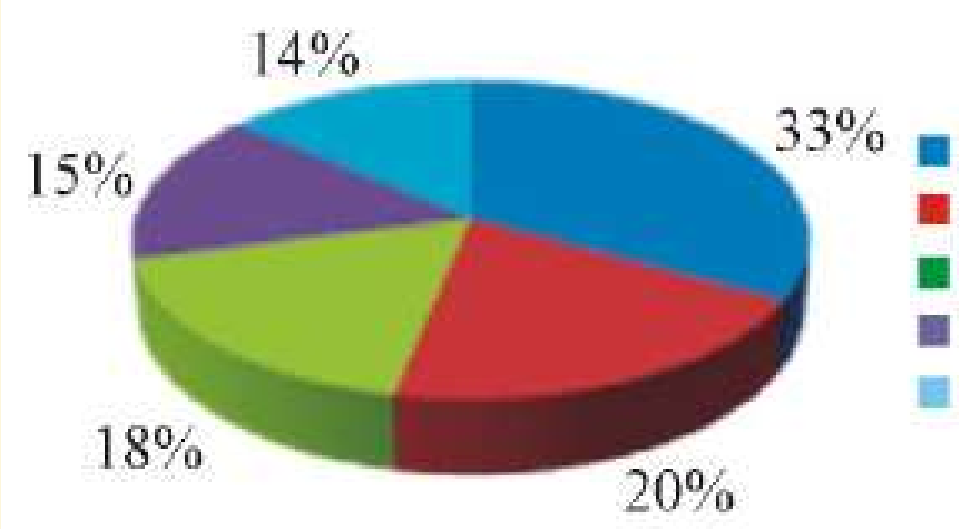
Tensile strength not up to standard



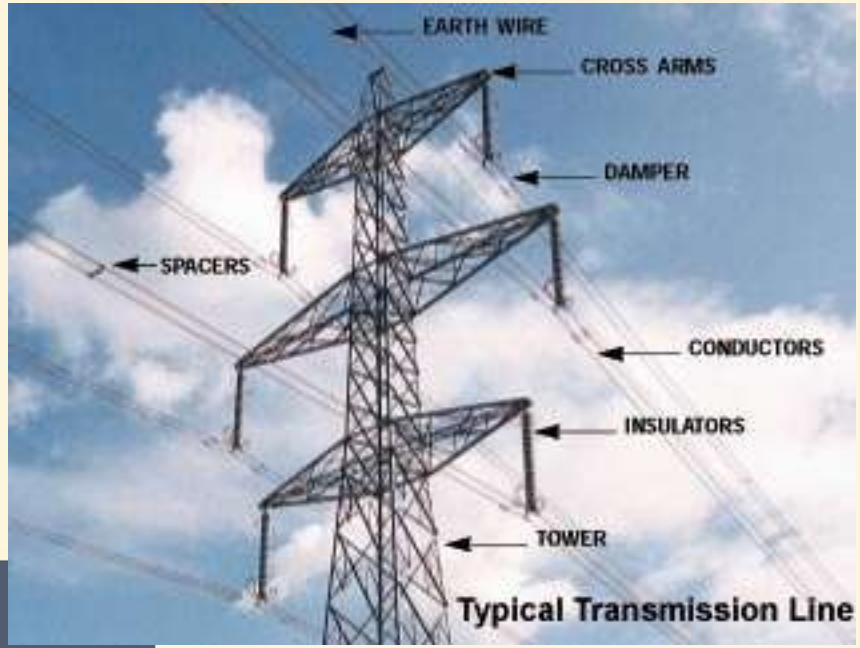


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Types of ***wire*** problems proportionally distributed



- Unsatisfactory mechanical properties
- Unsatisfactory resistance
- Wire break
- Wire corrosion
- Other disqualification

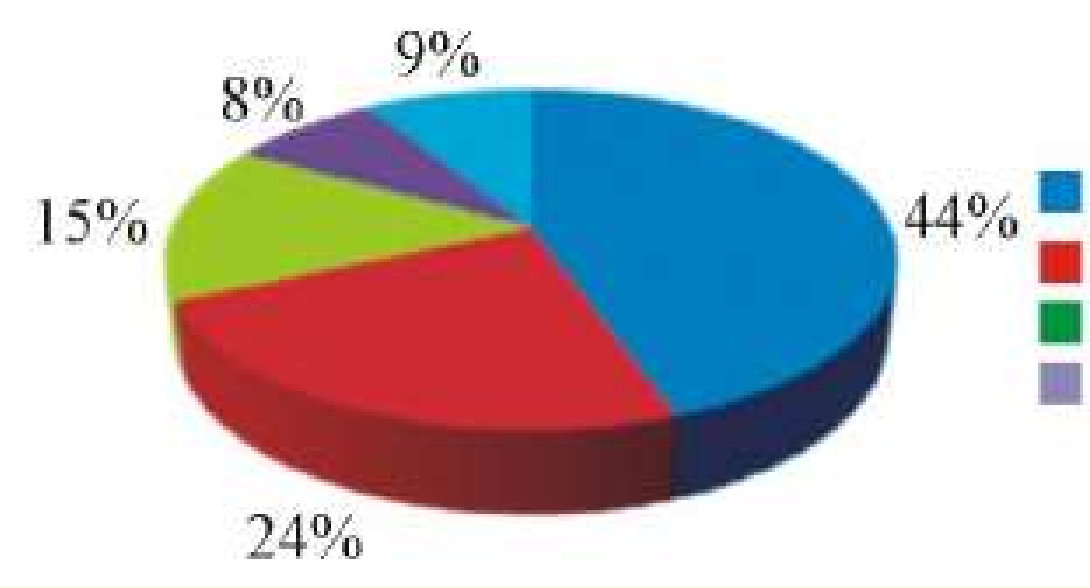




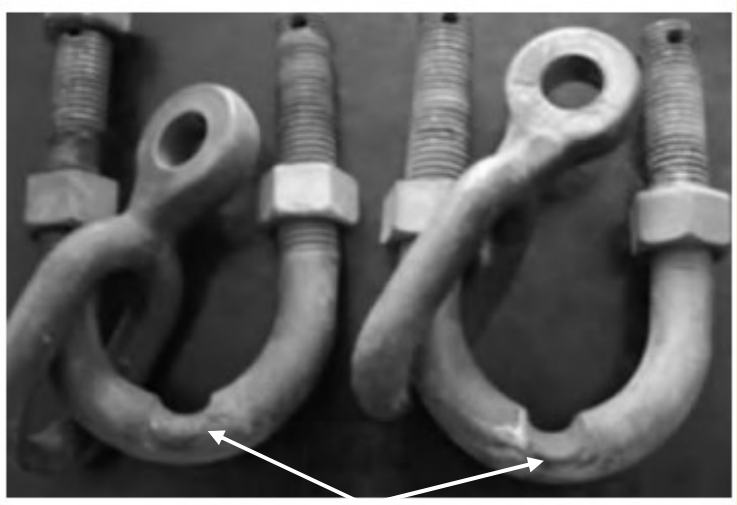
Wire corrosion

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Types of ***fitting*** problems proportionally distributed



- Unsatisfactory mech. properties
- Resistance test failed
- The fittings are unqualified
- Insulator bolt loose

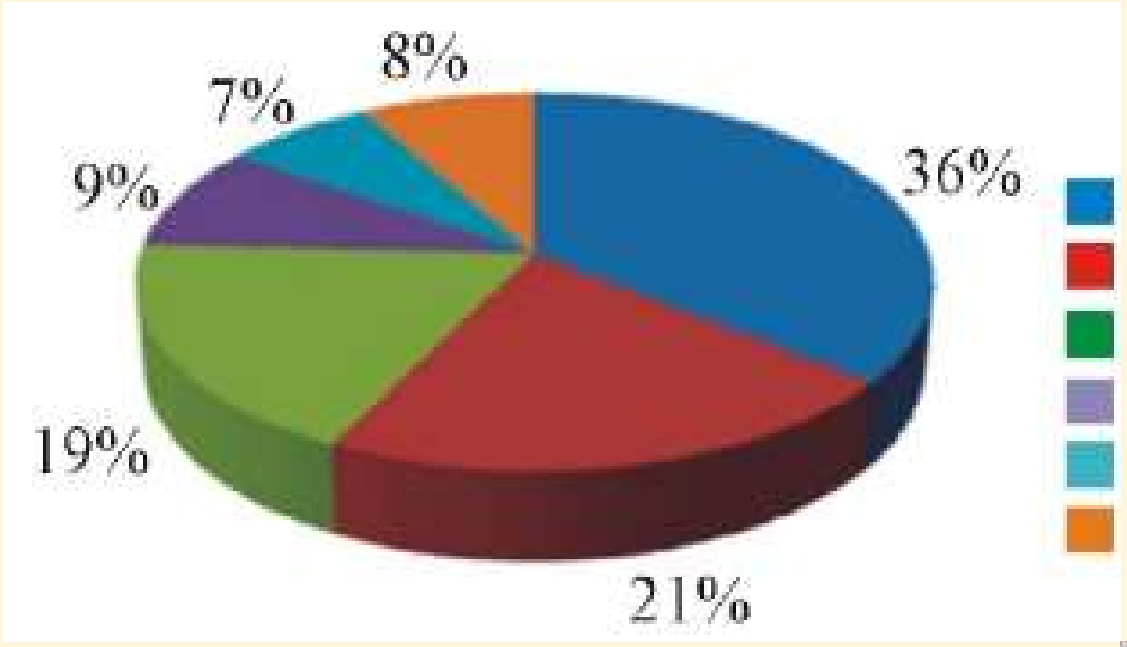


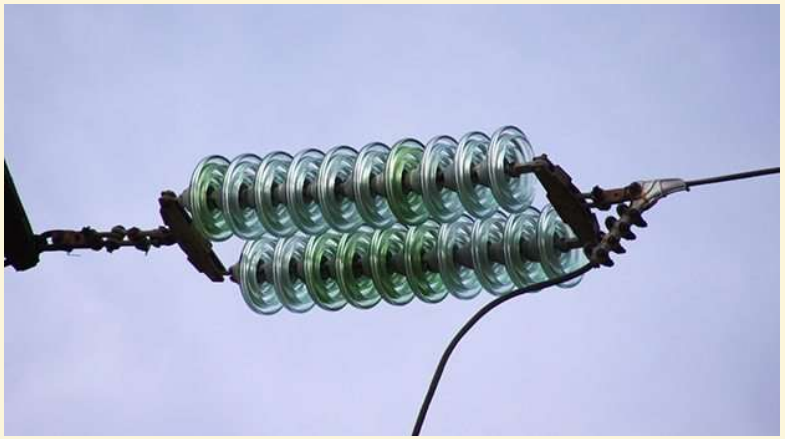
Fittings wear



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Types of ***insulator*** problems proportionally distributed







Insulator corrosion

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Corrosion failure of tower foot for hot galvanized steel transmission tower



Tower is located in the coastal area and village field.

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Corrosion failure of tower foot for hot galvanized steel transmission tower





Tower foot of 220kV line. Tower model 220Z1. Height 17.7m. Mass about 3t. Local corrosion is quite serious. Material has been significantly reduced and become thin.

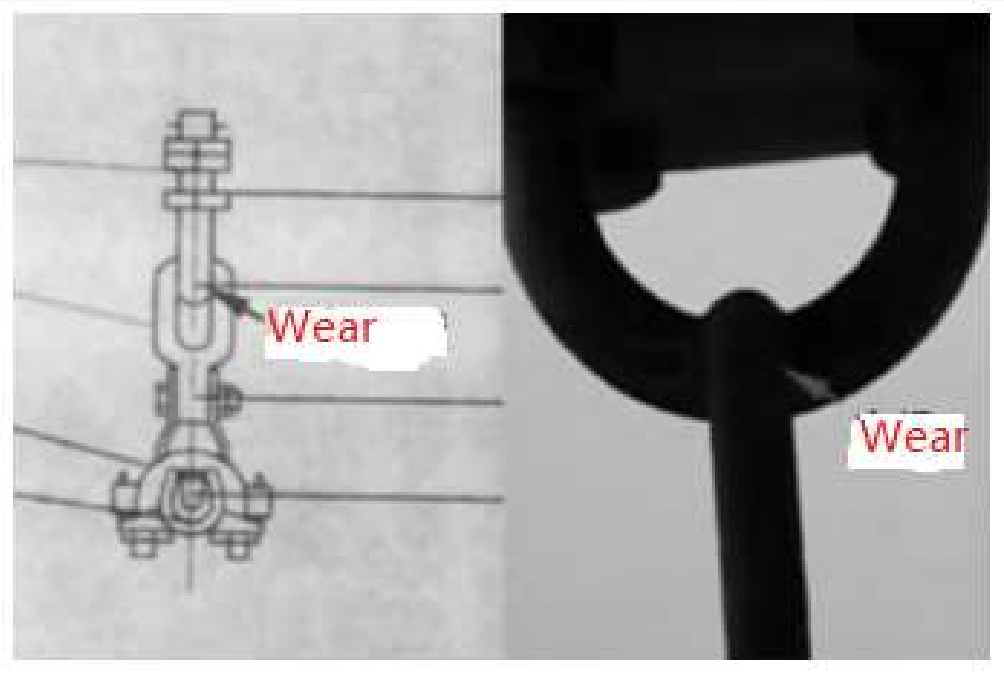
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Protective measures

- The principle is to interrupt the relevant links that are beneficial to the corrosion process of the tower feet.
- Apply a protective layer with a thickness of about 1 mm from the upper to lower 50 mm at the connection interface between the tower foot and the concrete foundation, which can isolate the corroded parts from the corrosive medium.
- Make the concrete protective cap beveled to prevent dust from accumulating in the root of the tower.
- Keep regular inspections and remove dust and salt accumulations in the tower feet timely.
- All measures above should have been applied to the towers of some transmission lines for more than one year.

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Wear of tower ground wire fittings

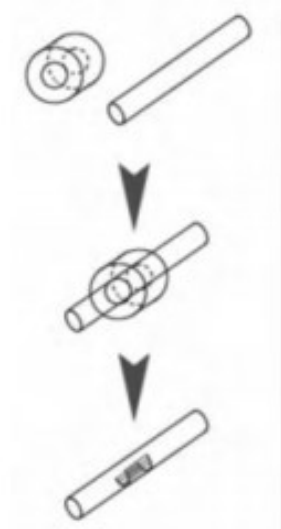


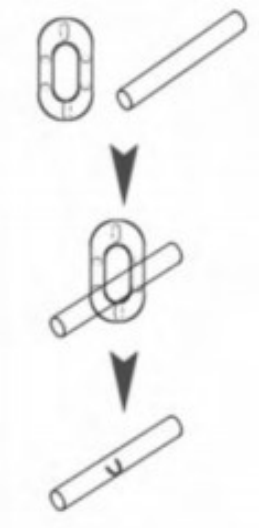


Tower ground wire fittings wear to form grooves.

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Wear of tower ground wire fittings





- The ***right angle*** hanging ring type hanging ring is subjected to the whole surface force, the contact force surface is large, and the movable space between the metal fittings is small, which reduces the collision wear between the two. The structure of the metal fitting is relatively stable and is not easy to wear.
- The ***annular type*** contact surface is small, the hanging ring is a point of force, and the movable space between the metal fittings is large, and the metal fitting structure has defects and is easy to wear.

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Wear test method for fittings of transmission line



U-7 hanging ring is worn.



Screw wear.

Chinese Standard Tests
GB/T 5075-2001, GB/T 2317.1-2008, GB/T 2314-2008,
DL/T 756-2 009, DL/T 759-2009

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Learning objectives

- Learn about the concepts of **fracture** and its importance in engineering design

- Become familiar with the main mechanisms of fracture, and their fundamental origins:
  - **Brittle (fast) fracture**
  - **Ductile (tearing) fracture**

- Appreciate **why materials fail prematurely** (i.e. before yielding) and how the **structural defects** in a material influence its fracture behaviour

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How do crystalline materials fail?

- **Plastic deformation**
  - Low temperature yielding
  - High temperature creep
- **Fracture**
  - Brittle fractures
  - Ductile fractures
- **Fatigue**
  - Slow crack growth leading to fracture
- **Oxidation** and **Corrosion**
  - Loss of material due to reactions with O_2, and $O_2 + H_2O$
- **Wear**
  - All of the above, with relative motion between parts!

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Fracture concepts

- Fracture:
  - **Initiation** and **rapid propagation** of cracks within a material
  - Structure **no longer sustains any applied loading**
  - **Often occurs at nominal stresses/strains below those that the material is expected to sustain – need to understand why this is!**

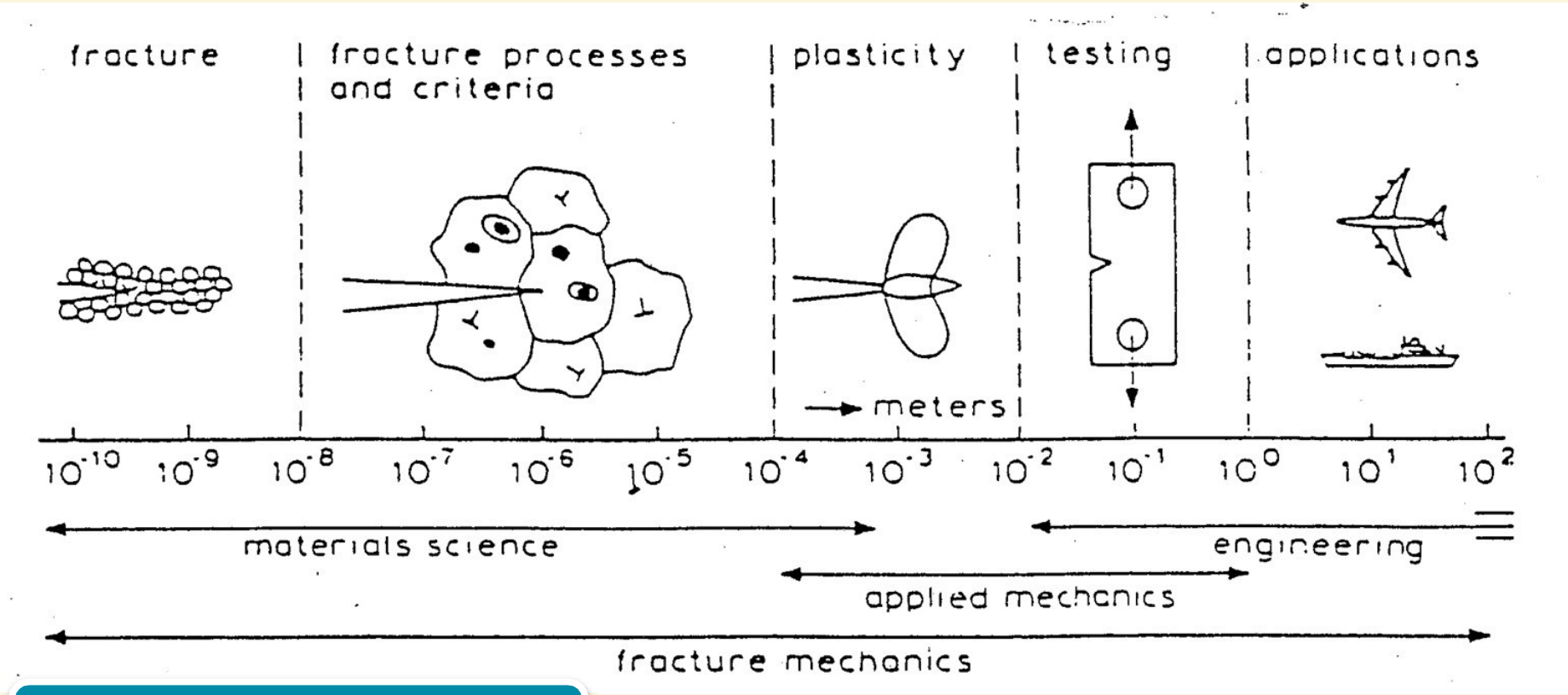
- Design against fracture:
  - Crack-free materials? **Very difficult to produce**
  - Therefore, we design for **non-propagation** or **controlled propagation** of cracks – this links in with fatigue behaviour
  - Need to select materials with sufficient **toughness**

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Fracture jargon

- Materials may be:
  - Strong
  - Stiff
  - **Tough**
  - **Ductile**
  - **Brittle**…

- Components may fail by:
  - **Inelastic deformation** – i.e. by yielding or creep
  - **Fracture** – i.e. brittle or ductile, or following fatigue
  - **Corrosion** – can cause failures independently or contribute to the above
  - ***Wear*** *– all failure mechanisms contribute to wear mechanisms*

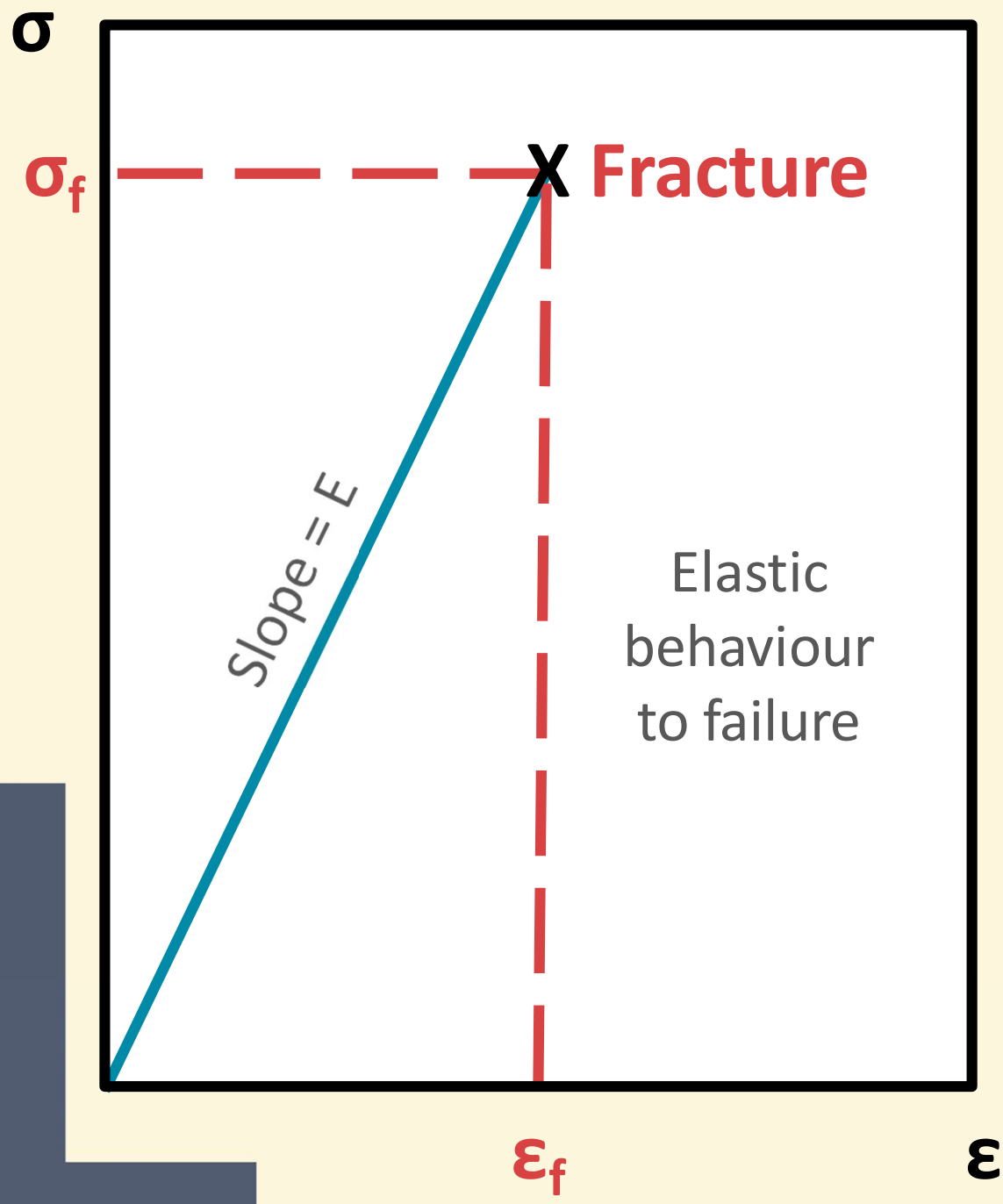


Our Course

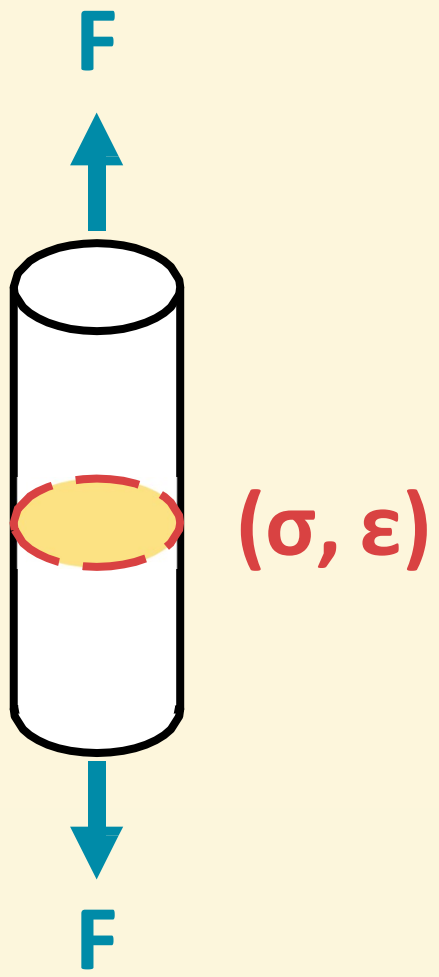
Fracture length scales

- Unlike many areas of engineering, fracture mechanisms operate on every length scale:
  - Atomistic to entire engineering systems

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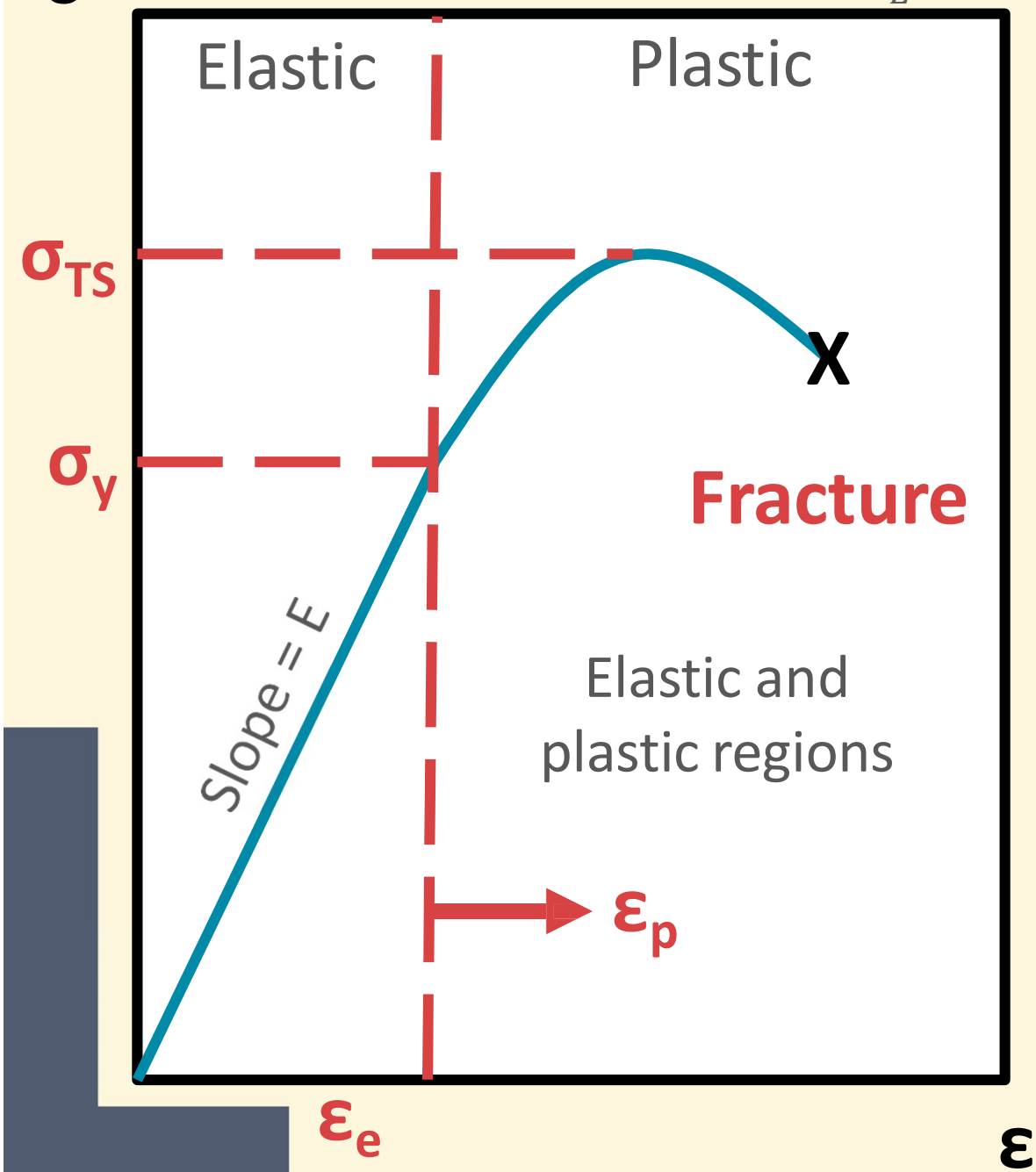


Deformation of brittle materials



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Total strain, $\varepsilon = \varepsilon_e + \varepsilon_p = \frac{\sigma_y}{E} + \varepsilon_p$



Deformation of ductile materials

